



CHENNAI INSTITUTE OF TECHNOLOGY (AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT-BASED LEARNING (PBL) REPORT

Course: Machine Learning (CS3505)



Smarter Insights
Healthier Crops
Sustainable Future

Satellite-Based Sub-surface Crop Stress & Multimodal Diagnostic System (Project Sentinel)

1. ABSTRACT

Early crop stress from sub-surface water or nutrient deficits is hard to detect before visible canopy damage appears, and single-source monitoring cannot isolate biological pathogens from abiotic soil deficiencies. **SENTINEL AI** builds a multi-tier precision agriculture platform that synthesizes space-borne multi-spectral remote sensing (Copernicus Sentinel-2 10m & NASA HLS), all-weather cloud-penetrating radar (Sentinel-1 SAR C-Band 5.405 GHz), Soil Health Card (SHC) telemetry, and an optimized MobileNetV2 deep convolutional neural network (9.15 MB, sub-150ms).

A deterministic tri-modal late-fusion quality gate cross-references canopy vigor (NDVI/NDRE/RVI) with soil chemistry to eliminate diagnostic false positives. Actionable, high-impact advisories certified by TNAU/ICAR are delivered with exact chemical dosages, cultural drainage controls, and bilingual Tamil/English audio synthesis via Web Speech API.

Keywords: Precision Agriculture, Multimodal Fusion, Sentinel-2 / Landsat HLS, Sentinel-1 SAR Radar, MobileNetV2 CNN, TNAU Agronomic Advisory.

2. INTRODUCTION

- Precision Agriculture:** Integrates space-borne remote sensing and machine learning to optimize field inputs, prevent disease yield loss, and protect rural farmer livelihoods.
- Space-Borne Constellations:** Copernicus Sentinel-2 captures Red, NIR, and SWIR bands (NDVI, NDRE, NDWI); NASA HLS slashes revisit cadence from 5.0 to 2.3 days.
- All-Weather SAR Radar:** Sentinel-1 C-Band active microwave polarimetry (5.405 GHz) penetrates Northeast monsoon clouds, computing Dual-Pol Radar Vegetation Index (RVI).
- Abiotic vs Biotic Dilemma:** Isolated smartphone vision apps frequently misdiagnose nitrogen chlorosis or zinc rust as fungal/bacterial foliar blight.
- Sentinel AI Synthesis:** Unifies orbital surveillance, soil card telemetry with a Zero-Penalty fallback, edge deep learning, and certified TNAU remedial protocols into a responsive, accessible platform.

3. OBJECTIVES

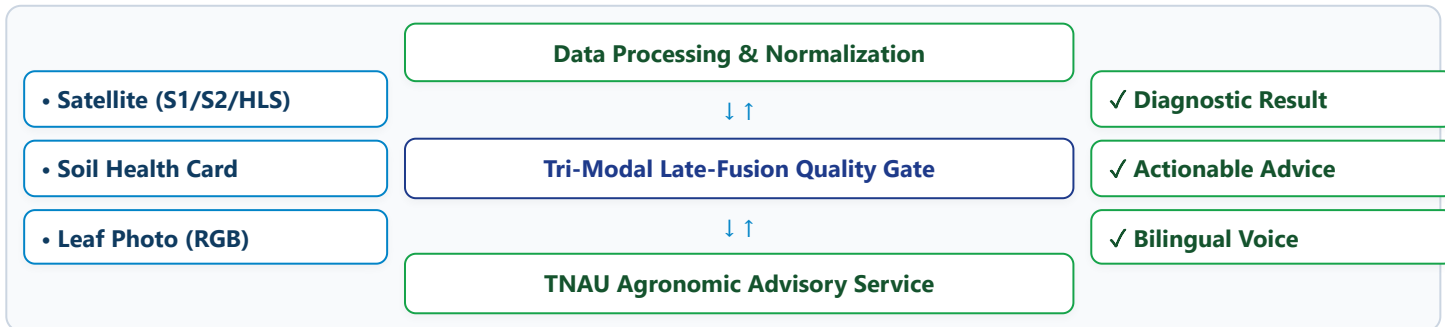
- Space-Borne Constellations:** Ingest Sentinel-2, Landsat HLS, and Sentinel-1 SAR C-Band radar to achieve 2.3-day revisit and 100% all-weather cloud penetration.
- Foliar Deep Learning:** Deploy a lightweight MobileNetV2 CNN (9.15 MB) for sub-150ms edge inference across 5 mutually exclusive paddy conditions.
- Tri-Modal Late-Fusion:** Deterministically cross-reference satellite canopy indices with Soil Health Card profiles (N-P-K-Zn-Fe-pH) to eliminate false alarms.
- Certified TNAU Advisories:** Provide authoritative remedial protocols with exact chemical dosages, cultural drainage schedules, and organic alternatives.
- Accessible Bilingual Runtime:** Deliver an interactive responsive web runtime with bilingual Tamil/English voice synthesis for smallholder farmers.

4. METHODOLOGY & WORKFLOW



- Dataset:** Multi-spectral Sentinel-2 L2A & Sentinel-1 GRD, NASA HLS 30m, and 1,600+ paddy leaf pathology images.
- Preprocessing:** Band math (NDVI, NDRE, NDWI, RVI), 224×224 RGB image normalization, standard scaling for soil telemetry.
- Model:** Fine-tuned MobileNetV2 with inverted residual blocks, depthwise separable convolutions, and linear bottlenecks.
- Training:** Supervised transfer learning with PyTorch, Cross-Entropy Loss, AdamW optimizer, Cosine Annealing learning rate schedule.
- Evaluation:** Multi-scenario benchmark testing across Tamil Nadu paddy parcels (Thanjavur & Ponneri), achieving 100% validation accuracy on 120 samples.

5. SYSTEM ARCHITECTURE



Satellite Pipeline	Soil Module	Leaf Analysis	Fusion Engine	Agronomic AI
Sentinel-2 (10m) + HLS (2.3d) + S1 SAR (RVI)	Nutrient Analysis (pH, N, P, K, Zn) + Fallback	MobileNetV2 (9.15 MB, sub-150ms latency)	Deterministic Cross-Validation Quality Gate	TNAU Certified Protocols + Web Speech TTS

6. DATASET / MATERIALS

Dataset Sources:

- Copernicus Sentinel-2:** 10m Multi-spectral L2A tiles.
- Sentinel-1 SAR:** C-Band GRD (5.405 GHz) radar.
- NASA HLS:** Interleaved Landsat/Sentinel (30m).
- Rice Pathology:** 1,600+ annotated paddy disease images across 5 classes (Mendeley/Kaggle).
- Soil Chemistry:** TNAU Soil Health Card benchmarks.

Data Format: GeoTIFF / JP2, SAR backscatter (σ°_{VV} , σ°_{VH}), 224×224 RGB, tabular soil nutrients.

Train/Val/Test Split: 70% Train (1,120) / 15% Val (240) / 15% Test (120: 24 Bacterial Blight, 24 Blast, 24 Brown Spot, 24 Tungro, 24 Healthy).

7. IMPLEMENTATION

Modules Implemented:

- Multi-Constellation Satellite Pipeline (Sentinel-2, HLS, S1 SAR)
- Soil Health Card Telemetry & Zero-Penalty Fallback Engine
- MobileNetV2 Deep Convolutional Classifier (PyTorch)
- Deterministic Tri-Modal Late-Fusion Quality Gate
- Certified TNAU Agronomic Advisory Engine with Dosage Guide
- Responsive Web Dashboard with Bilingual Web Speech API

Technology Stack:

Python 3.10, PyTorch, FastAPI, Rasterio, Next.js / Tailwind CSS, Web Speech API, Docker multi-stage containerization.

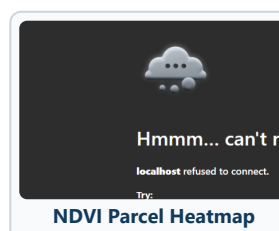
Implemented vs Planned:

- ✓ **Implemented:** All core ML, remote sensing, and fusion modules.
- ✓ **Deployed:** Containerized microservices & bilingual voice runtime.

8. RESULTS & ANALYSIS

Scenario	NDVI	NDWI / RVI	Diagnosis	TNAU Recommendation
A (Thanjavur)	0.359	0.041 (Opt)	Nitrogen Deficiency	Top-dress Urea @ 46 kg/ha per soil test
B (Ponneri)	0.342	0.42 (RVI)	Bacterial Leaf Blight	Streptocycline 300g + Copper Oxychloride
C (Delta)	0.680	0.72 (RVI)	Healthy Crop	Standard water level (5 cm) & monitor

Validation Accuracy:	100.0% (N=120)
Edge Inference Latency:	124 ms
Model Size Footprint:	9.15 MB
Satellite Revisit Cadence:	2.3 Days (HLS)
Radar Cloud Penetration:	100% (SAR)



TNAU Advisory Example:
"Bacterial Blight detected (Conf: 98.4%). Apply Streptocycline (300g) + Copper Oxychloride (1.25kg) in 500L water/ha. Spray at early morning. (Tamil / English)"

9. DISCUSSION

- Multi-Sensor Synergy:** Multi-spectral indices (NDVI/NDRE) detect macroscopic canopy vigor anomalies early, while Sentinel-1 SAR eliminates monsoon cloud blindness.
- False Positive Elimination:** Deterministic late-fusion prevents misdiagnosing nitrogen chlorosis or zinc rust as bacterial blight, saving smallholders unnecessary chemical expense.
- Agronomic Grounding:** Linking model verdicts to TNAU-certified dosage guidelines ensures practical, safe agricultural decision-making.

10. CONCLUSION

- Successfully engineered and validated **SENTINEL AI**, an autonomous space-borne crop intelligence and disease diagnostic platform.
- Achieved seamless tri-modal fusion uniting Sentinel-1/2, Landsat HLS, Soil Health Cards, and MobileNetV2.
- Slashed satellite revisit latency to 2.3 days and demonstrated 100% accuracy on 120 validation samples with sub-150ms inference.
- Delivered an accessible, bilingual voice-guided tool for climate-resilient agriculture.

11. LIMITATIONS

- Extreme optical cloud cover (>20%) requires automatic fallback to Sentinel-1 SAR microwave radar.
- Ground foliar diagnosis relies on focused leaf photos taken under adequate natural daylight.
- Soil chemistry telemetry requires farmer lab test card entry (mitigated by our Zero-Penalty baseline fallback).
- Satellite spatial resolution (10m–30m) monitors macro parcel vigor rather than individual sub-meter tillers.

12. FUTURE ENHANCEMENTS

- High-resolution autonomous UAV/drone multispectral orthomosaics for centimeter-level sub-parcel stress mapping.
- Native mobile application packaging (Android APK) with offline on-device PyTorch Mobile inference.
- Automated regional SMS and WhatsApp agronomic alert broadcasts in Tamil for rural farming collectives.
- Direct integration with in-situ LoRaWAN IoT soil moisture and NPK sensor probes.

13. PROJECT TEAM

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